The Systematic Investor Report

Recent market turmoil – what happened to liquid alternatives?

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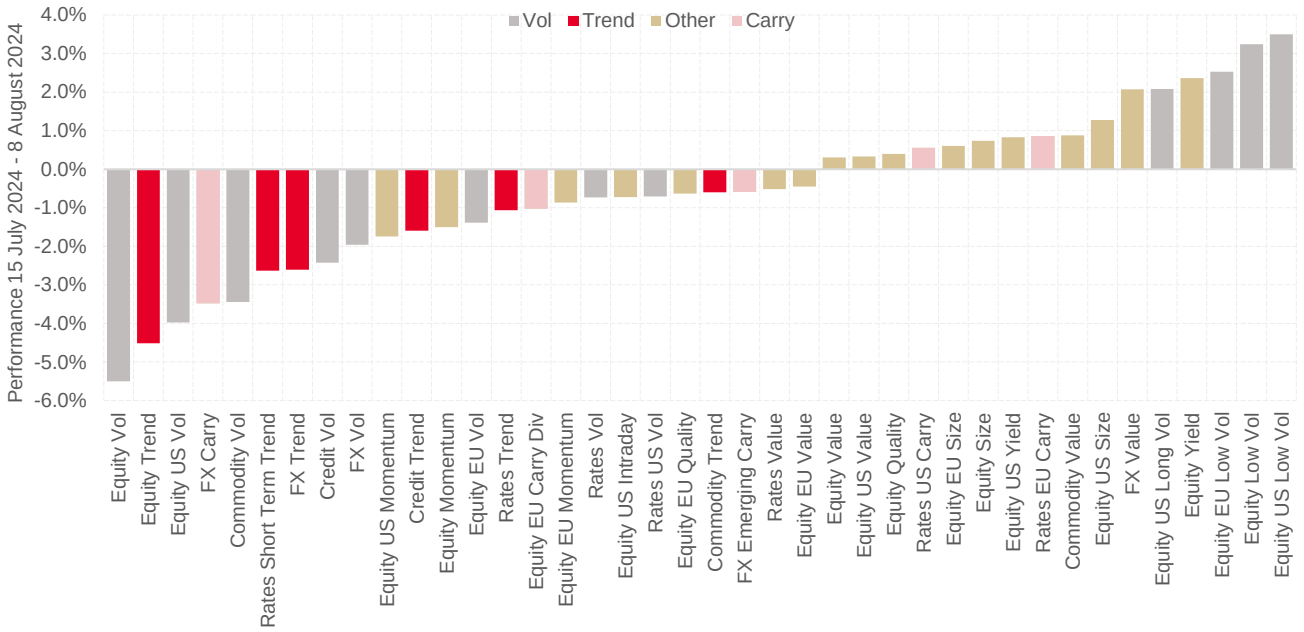
- Monday 5 August saw the second-biggest daily rise in VIX and Tuesday 6 saw the second-sharpest decline since the "Volmaggedon" of February 2018. This percentage move was partly due to a sudden repricing of the left tail, which now accounts for 30% of the VIX. The unwinding of the Yen carry trade and its impact on Japanese equities was one of the major catalysts of the sell-off. VNKY reached 70%, Nikkei 225 sold-off by 12.5%, the SG risk sentiment indicator reached extreme risk aversion levels below 0.05, and liquidity fell sharply in listed markets.

- The performance of quantitative strategies over the past few weeks has been mostly typical of a "risk-off" event. Strategies that short volatility or harvest carry from risk assets have been hit quite badly, with FX carry and buying European dividend futures among the worst performers.

- Trend following also had a tough time. The SG Trend Index lost most of this year's gains. The repricing of the growth and interest rate outlook at the end of June, coupled with the most recent sell-off created a sharp reversal, first in commodities prices, and then in equities. Given the current positioning, trend followers remain exposed to a continuation of the sell-off in equities but should be fine in a rebound. They have just turned long duration on major bond markets, and are positioned for a recessionary scenario in commodities, with a short on agricultural commodities and a long in gold prices (see page 3 for more on CTAs positioning).

- Long volatility strategies in options benefited from the sell-off as well as low volatility stocks versus high volatility stocks. The odd one out is probably the performance of the size factor in the US and in Japan. The rotation from large to small was aggressive in July and extended until early August. This is one of the consequences of the AI stock bubble starting to deflate. Tail hedges have started to kick in and are now better positioned for a continuation of the sell-off. In the following pages, we discuss the benefits of diversifying tail hedging, and how a diversified portfolio of tail hedges would have performed. Lastly, after a volatility peak, carry opportunities increase. We update our carry monitor on page 5 of this note.

Performance of quant factors since July 2015



Source: SG Cross Asset Research SGI, Bloomberg

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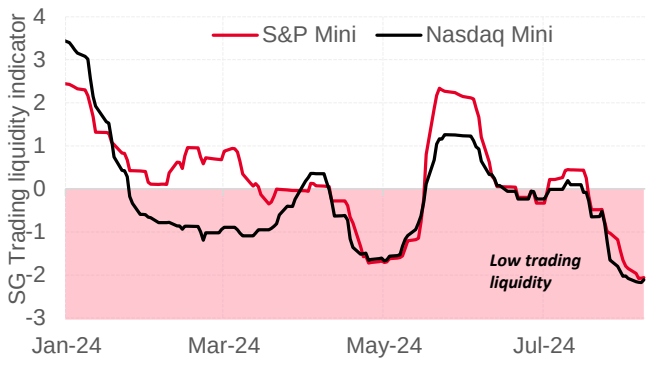
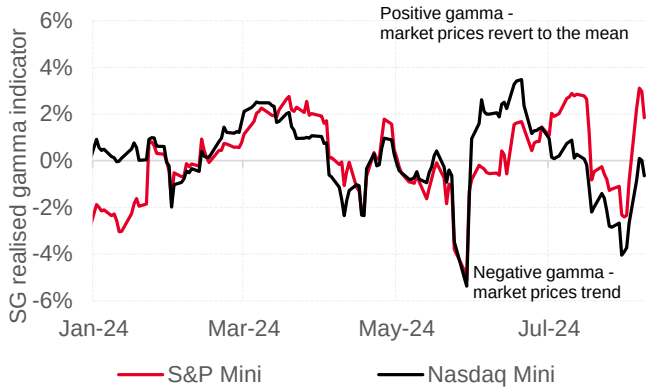
Tail hedges in the storm

Heightened risk aversion is creating difficult trading conditions with very low levels of available liquidity. The SG trading liquidity indicator has dropped into the danger zone. This is generally good news for intraday trend followers. Moreover, our realised gamma indicator has turned positive for the S&P, and neutral for the Nasdaq, hinting that intraday trend following would be better in the Nasdaq.

Liquidity indicators in the danger zone



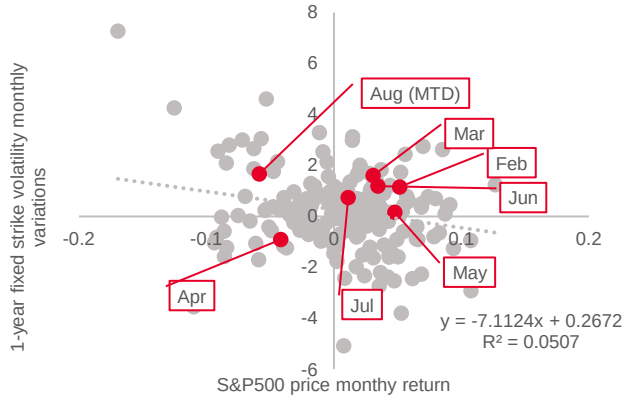
Gamma hedgers have turned long vol for the S&P



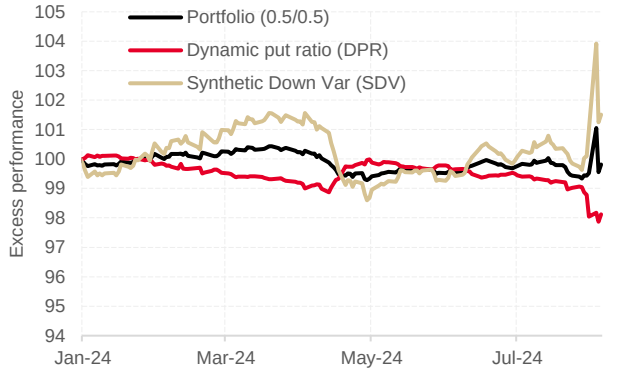
Source: SG Cross Asset Research/Cross Asset Quant

When liquidity conditions deteriorate, the probability of an accident increases, and therefore tail hedges become more relevant. We discussed two tail hedges in our June outlook: Synthetic Down Variance (SDV) and Dynamic Put Ratio (DPR). We argued for a diversified approach combining these two strategies. The past week demonstrated the benefits of this approach. As volatility rose Synthetic Down Variance posted a 3% gain on Monday when the S&P was down 3%. Dynamic Put Ratio was up 0.13% on the same day, after having lost money the week before.

Spot/vol dynamics – August is spot down/vol down so far



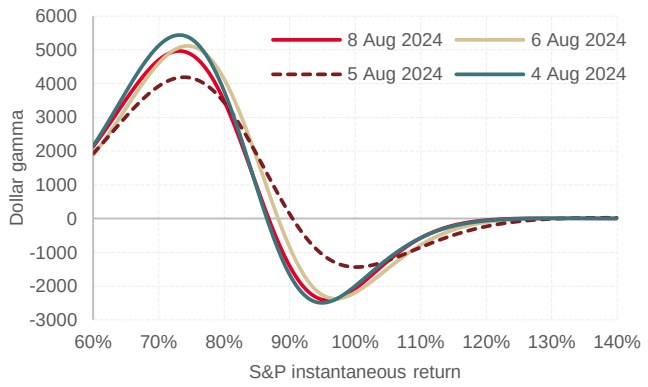
Combining tail hedges



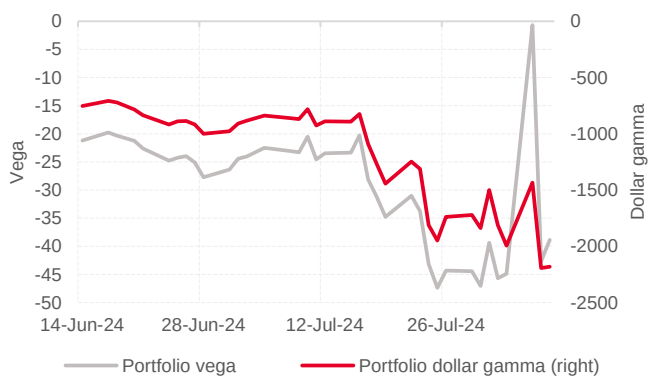
Source: SG Cross Asset Research/Cross Asset Quant

The performance of Dynamic Put Ratio over the period is in line with expectations. Outside of a tail event, the gamma of Dynamic Put Ratio is negative and its vega is negative too. The net gamma of the strategy moved closer to its trough on Monday. Going forward, the strategy remains positioned for an extension of the sell-off. And if the market rebounds, the trade should benefit in the event that realised volatility is lower than implied volatility.

Gamma profile in DPR – the strategy is close to its gamma trough…



…and it has now returned to a negative vega position



Source: SG Cross Asset Research/Cross Asset Quant

What's next for CTAs?

Timing CTA/Trend strategies is as complicated as timing markets. The SG Trend Index lost most of this year's gains. After such a large drop in the strategy performance, it is fair to look at the positioning and try to understand the performance going forward.

In currencies and equities, the worst performance came from CTAs' long positions in Nikkei 225, S&P500 and JPY. According to our model, they have now switched to a neutral position on JPY but are still long on most equity benchmarks. If the market continues its rebound, they should be able to recover some of their losses.

Equity and currency estimated positioning for CTAs

	Current Position	Quantile (5y)	Days in Position	Monthly Perf.	YTD Perf.
JPY/USD	Flat		1	-7.4%	2.9%
CHF/USD	Flat		6	-2.4%	-6.4%
GBP/USD	Long	56%	19	-1.6%	-3.4%
EUR/USD	Flat		10	-1.1%	-6.2%
AUD/USD	Short	69%	5	-0.8%	-5.4%
SEK/USD	Short	32%	14	-0.3%	-4.4%
NZD/USD	Short	66%	16	1.1%	-4.8%
NOK/USD	Short	49%	16	1.9%	0.9%
CAD/USD	Short	69%	9	2.2%	-0.4%
	Current Position	**Quantile (5y)**	**Days in Position**	**Monthly Perf.**	**YTD Perf.**
NKY	Long	24%	27	-7.9%	-0.3%
S&P	Long	59%	140	-6.7%	4.1%
AEX	Long	32%	136	-5.4%	4.0%
IBEX	Long	58%	18	-5.2%	1.1%
ASX	Long	62%	15	-3.6%	-4.8%
FTSE UK	Long	72%	5	-2.9%	-2.4%
SX5E	Flat		6	-1.8%	1.8%

Source: SG Cross Asset Research/Cross Asset Quant

The story in commodities and bonds is very much in line with a recessionary scenario: flat to short positions in commodities linked to the real economy whilst being long on gold and government bonds. Interestingly, the switch from being short to long on bonds happened this week, on the back of the more dovish comments from the Bank of Japan and the Fed rates cut repricing.

Commodities and bonds estimated positioning for CTAs

	Current Position	Quantile (5y)	Days in Position	Monthly Perf.	YTD Perf.
Copper	Flat		7	-3.4%	-2.9%
Zinc	Flat		8	-3.0%	-8.1%
Silver	Long	28%	4	-3.0%	1.7%
Aluminum	Flat		1	-2.3%	-1.9%
Crude Oil	Flat		1	-1.4%	-0.6%
Heating Oil	Short	46%	9	-0.9%	-3.1%
Gold	Long	58%	116	0.8%	10.9%
Cotton	Short	46%	9	1.4%	0.6%
Natural Gas	Short	52%	26	1.6%	1.8%
Corn	Short	39%	43	1.9%	10.9%
Wheat	Short	48%	32	2.3%	-1.0%
Sugar	Short	33%	20	2.5%	0.5%
Soybean	Short	37%	41	5.7%	9.8%
	Current Position	**Quantile (5y)**	**Days in Position**	**Monthly Perf.**	**YTD Perf.**
France	Long	23%	1	-1.4%	-3.3%
Japan	Short	27%	83	-3.0%	-2.0%
Germany	Long	21%	3	0.5%	-4.9%
Australia	Long	30%	1	-2.2%	-6.8%
US	Long	26%	1	1.2%	-3.6%
Canada	Long	41%	4	2.1%	-3.8%
Italie	Long	53%	16	0.4%	-2.3%
UK	Long	41%	4	2.9%	-5.2%

Source: SG Cross Asset Research/Cross Asset Quant

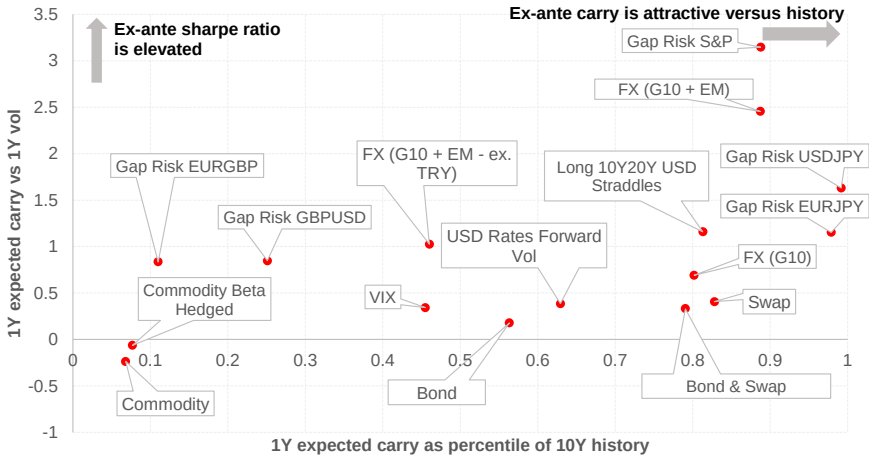
Overall, the positioning of CTAs is quite balanced, with equities and currencies still positioned for a rebound, and rates and commodities defensively positioned.

Exploring carry oportunities

Lastly, the graph below is an update of our carry monitor. A few trades worth highlighting:

- A strategy monetising gap risk offers interesting characteristics either in equity (S&P and Nasdaq) or in currencies (JPY pairs).
- The FX carry strategy remains attractive when adding extremely high yield currencies such as the Turkish Lira. Our EM strategists remain positive on the outlook for this currency.
- Buying straddles on USD 20y rates for a 10-year expiry remains an attractive proposal from a carry standpoint.
- VIX carry is evolving favourably, with an improved entry point and is something to watch out for in the coming weeks.

After a volatility peak, carry opportunities increase



Source: SG Cross Asset Research/Cross Asset Quant

Equity performance analysis

Fundamental premia

	Monthly return	Return YTD	Volatility YTD	Return/risk YTD	Max drawdown since 2015	Bench return/risk YTD	Correl with Bench	Correl with MSCI World	Correl with 10Y US rates	Tail correl with MSCI World	Tail correl with 10Y US rates
Value World	-0.7%	0.1%	3.5%	0.0	-25.5%	-0.1	75.8%	-21.5%	35.0%	18.5%	37.8%
Value Europe	-3.5%	-0.5%	4.1%	-0.1	-27.1%	-0.4	77.5%	5.2%	36.0%	40.7%	49.7%
Value US	2.1%	-0.6%	4.4%	-0.1	-29.6%	-0.4	90.0%	6.8%	-3.9%	55.1%	33.8%
Quality World	1.6%	1.8%	3.7%	0.5	-18.1%	0.9	27.1%	-37.5%	18.0%	-72.8%	-32.4%
Quality Europe	1.4%	2.9%	3.6%	0.8	-17.9%	0.8	17.8%	-34.8%	16.7%	-69.7%	-39.6%
Quality US	-1.8%	2.4%	4.5%	0.5	-26.4%	0.7	-3.5%	-21.0%	19.8%	-71.3%	-30.5%
Profitability World	-0.3%	1.6%	3.8%	0.4	-9.7%	0.4	75.1%	11.1%	9.5%	0.9%	-19.9%
Profitability Europe	-0.6%	-0.3%	3.1%	-0.1	-13.9%	-1.1	58.0%	-16.9%	18.8%	-32.2%	-44.4%
Profitability US	-1.7%	-0.8%	3.4%	-0.2	-14.7%	0.3	84.6%	8.3%	8.0%	-44.1%	-37.8%
Carry World	1.6%	-0.6%	4.0%	-0.1	-28.2%	-0.4	30.8%	-25.0%	-5.2%	3.6%	14.3%
Carry Europe	-0.6%	3.3%	3.5%	0.9	-31.1%	0.6	8.3%	-26.5%	34.7%	-13.9%	30.7%
Carry US	5.2%	1.9%	5.9%	0.3	-28.6%	-0.3	47.7%	-16.7%	-30.9%	-31.4%	-26.9%
Size World	0.3%	-6.5%	3.4%	-1.9	-30.8%	-1.6	28.2%	-2.2%	-26.9%	-21.3%	-20.6%
Size Europe	-0.6%	-0.1%	3.0%	0.0	-17.1%	-1.4	45.1%	48.7%	-38.2%	47.1%	6.7%
Size US	2.0%	-4.7%	4.1%	-1.1	-31.8%	-1.5	73.4%	28.0%	-33.8%	60.5%	30.0%
Low Vol World	3.2%	1.4%	4.0%	0.4	-25.4%	0.9	75.2%	-66.7%	20.7%	-77.8%	-28.9%
Low Vol Europe	2.3%	1.7%	4.0%	0.4	-25.0%	0.8	67.6%	-62.0%	27.9%	-80.6%	-29.8%
Low Vol US	3.0%	2.8%	5.1%	0.5	-31.1%	0.7	68.4%	-65.8%	25.6%	-79.2%	-30.5%
Momentum World	-1.5%	0.7%	3.4%	0.2	-15.6%			41.0%	3.5%	-22.9%	-7.5%
Momentum Europe	0.6%	2.0%	2.9%	0.7	-11.7%	1.9	39.3%	13.8%	-2.7%	-12.3%	-6.0%
Momentum US	-1.6%	3.1%	5.1%	0.6	-30.2%	1.6	67.4%	29.3%	12.8%	-25.6%	-4.7%
Trend World	-6.9%	5.9%	9.6%	0.6	-47.1%	1.8	17.5%	55.5%	-12.6%	-7.3%	18.2%

Source: SG Cross Asset Research/Cross Asset Quant. Correlations are calculated on a one-year rolling window. Bench refers to PremiaLab Pure Factor Indices. See addendum for more details.

Systematic hedging

	Monthly return	Return YTD	Volatility YTD	Return/risk YTD	Max drawdown since 2015	Bench return/risk YTD	Correl with Bench	Correl with MSCI World	Correl with 10Y US rates	Tail correl with MSCI World	Tail correl with 10Y US rates
Synthetic Down Var US	1.4%	1.7%	4.8%	0.3	-26.7%			39.8%	-41.0%	2.7%	-37.3%

Source: SG Cross Asset Research/Cross Asset Quant. Correlations are calculated on a one-year rolling window. Bench refers to PremiaLab Pure Factor Indices. See addendum for more details.

Trading premia

	Monthly return	Return YTD	Volatility YTD	Return/risk YTD	Max drawdown since 2015	Bench return/risk YTD	Correl with Bench	Correl with MSCI World	Correl with 10Y US rates	Tail correl with MSCI World	Tail correl with 10Y US rates
S&P Daily-Biweekly Mean Reversion	1.8%	-0.6%	11.2%	0.0	-32.5%			4.0%	14.9%	49.2%	5.3%
Machine Learning World	0.0%	0.0%	0.0%		-6.6%					-0.1%	-16.6%
Machine Learning Europe	-1.4%	-2.0%	4.3%	-0.5	-10.9%			-24.4%	24.7%	-1.6%	-1.5%
Machine Learning US	0.5%	6.0%	4.7%	1.3	-8.8%			-30.1%	31.9%	-7.5%	-12.1%
Machine Learning Japan	0.0%	0.0%	0.0%		-14.5%					6.5%	-9.7%
Intraday Trend Following Long	-1.9%	-3.3%	2.3%	-1.5	-16.8%			29.8%	-6.0%	2.2%	-5.9%
Intraday Trend Following Short	0.6%	-0.1%	1.0%	-0.1	-7.0%			-20.3%	-0.5%	-36.6%	0.4%

Source: SG Cross Asset Research/Cross Asset Quant. Correlations are calculated on a one-year rolling window. Bench refers to PremiaLab Pure Factor Indices. See addendum for more details.

Rates performance analysis

Fundamental premia

	Monthly return	Return YTD	Volatility YTD	Return/ risk YTD	Max drawdown since 2015	Bench return/ risk YTD	Correl with Bench	Correl with MSCI World	Correl with 10Y US rates	Tail correl with MSCI World	Tail correl with 10Y US rates
Bond Carry World	-0.5%	-0.2%	2.8%	-0.1	-12.6%	0.4	76.1%	-0.2%	37.4%	-18.4%	-22.2%
Swap Carry World	0.0%	3.1%	3.4%	0.9	-14.6%			-34.9%	75.3%	-13.5%	-18.9%
Trend World	-4.2%	-13.6%	4.5%	-3.0	-29.9%	-1.1	37.2%	-44.4%	52.6%	-49.0%	-10.2%
Volatility Europe	*0.0%*	*0.0%*	*0.0%*		*-28.6%*	*1.0*				*42.2%*	*-3.1%*
Value World						0.4					

Source: SG Cross Asset Research/Cross Asset Quant. Correlations are calculated on a one-year rolling window. Bench refers to PremiaLab Pure Factor Indices. . See addendum for more details.

Foreign exchange performance analysis

Fundamental premia

	Monthly return	Return YTD	Volatility YTD	Return/ risk YTD	Max drawdown since 2015	Bench return/ risk YTD	Correl with Bench	Correl with MSCI World	Correl with 10Y US rates	Tail correl with MSCI World	Tail correl with 10Y US rates
Carry World	-4.9%	3.4%	4.6%	0.7	-12.9%	0.7	94.1%	19.2%	42.4%	67.1%	47.2%
Value World	5.6%	2.7%	3.6%	0.8	-10.8%			-10.2%	-30.0%	-4.6%	-17.5%
Trend World	-3.0%	-7.8%	5.7%	-1.4	-34.0%	-1.7	73.3%	-44.1%	41.8%	-34.5%	27.4%
Volatility World	0.0%	0.0%	0.0%		-27.5%			-19.8%	13.4%	50.0%	25.6%

Source: SG Cross Asset Research/Cross Asset Quant. Correlations are calculated on a one-year rolling window. Bench refers to PremiaLab Pure Factor Indices. . See addendum for more details.

Credit performance analysis

Fundamental premia

	Monthly return	Return YTD	Volatility YTD	Return/ risk YTD	Max drawdown since 2015	Bench return/ risk YTD	Correl with Bench	Correl with MSCI World	Correl with 10Y US rates	Tail correl with MSCI World	Tail correl with 10Y US rates
Carry World	0.4%	-1.5%	1.4%	-1.1	-24.3%	-0.1	71.5%	29.9%	-38.2%	29.8%	2.0%
Carry Europe	*0.1%*	*-2.6%*	*1.9%*	*-1.3*	*-24.8%*			*18.6%*	*-30.9%*	*11.8%*	*8.5%*
Carry US	*0.6%*	*-0.4%*	*1.4%*	*-0.3*	*-27.6%*			*34.9%*	*-35.7%*	*40.1%*	*-0.6%*
Trend World	0.0%	0.0%	0.0%		-5.7%	0.0	50.4%	38.2%	-30.6%	-20.9%	0.7%

Source: SG Cross Asset Research/Cross Asset Quant. Correlations are calculated on a one-year rolling window. Bench refers to PremiaLab Pure Factor Indices. See addendum for more details.

Commodity performance analysis

Fundamental premia

	Monthly return	Return YTD	Volatility YTD	Return/ risk YTD	Max drawdown since 2015	Bench return/ risk YTD	Correl with Bench	Correl with MSCI World	Correl with 10Y US rates	Tail correl with MSCI World	Tail correl with 10Y US rates
Carry CODA World	0.7%	1.7%	1.3%	1.3	-3.4%	0.5	87.8%	-4.0%	-9.8%	-24.6%	-3.4%
Carry Optimix World	0.6%	0.4%	2.0%	0.2	-6.5%			-13.6%	0.0%	-23.3%	-2.6%
Momentum World	-0.5%	2.9%	6.1%	0.5	-25.1%	0.0	87.9%	-8.3%	19.1%	-30.1%	1.0%

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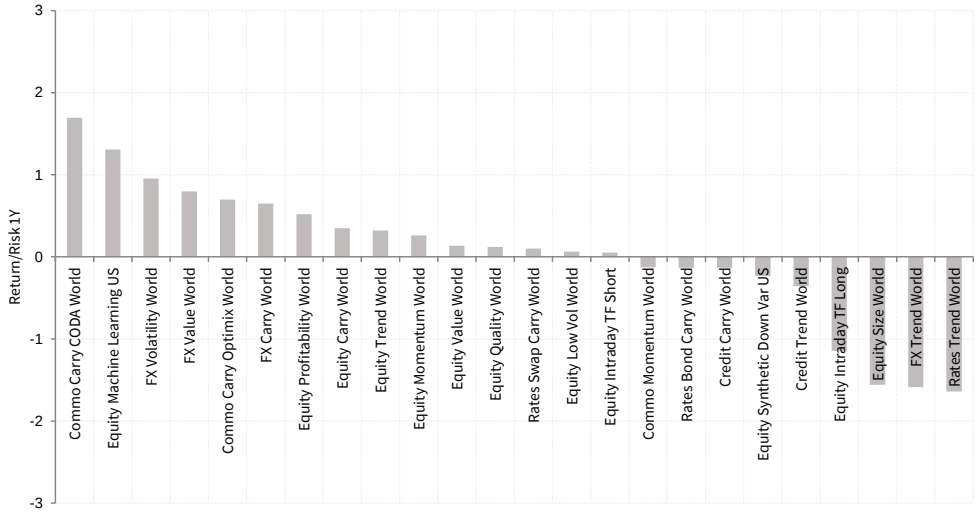
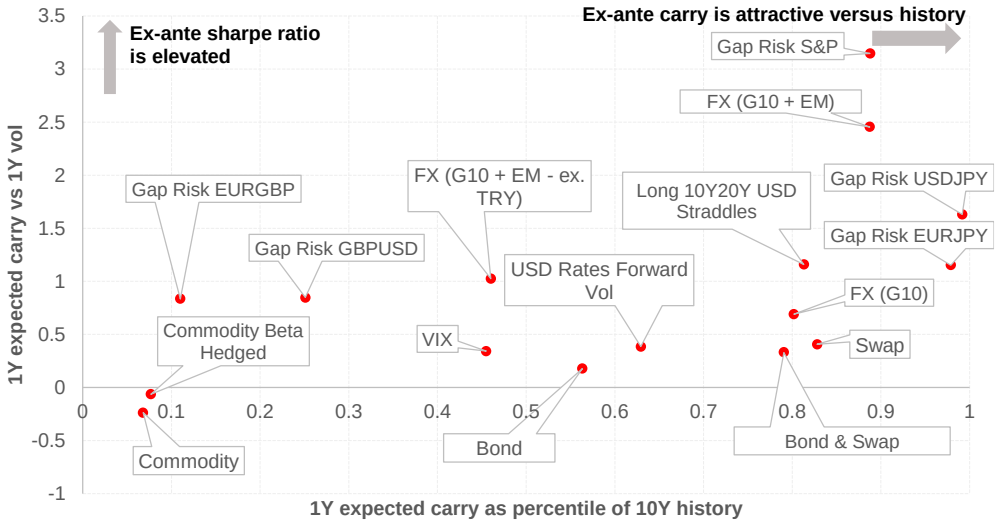
Top and bottom performers, 1y



Source: SG Cross Asset Research/Cross Asset Quant. Past performance is not indicative of future performance.

Carry monitor



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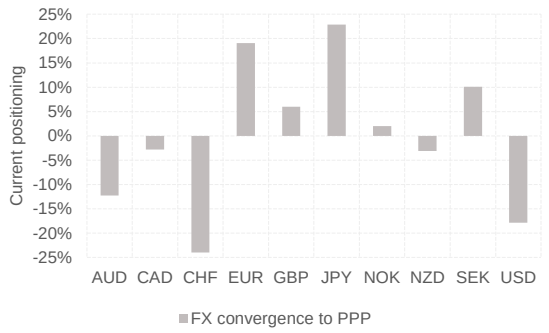
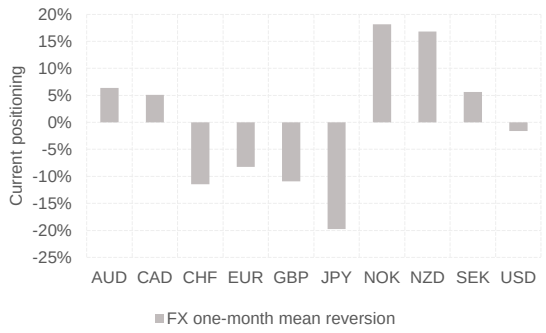
FX value positioning

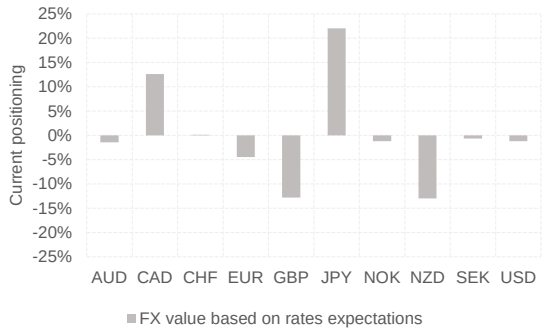
Convergence to PPP model current positioning



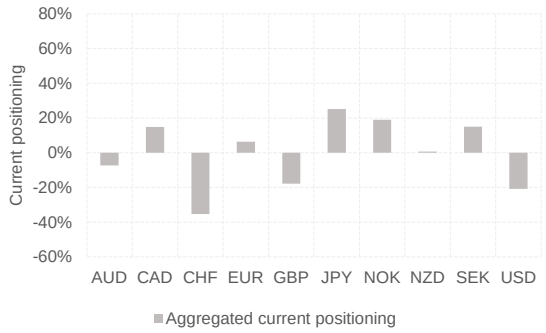
One-month mean-reversion model current positioning



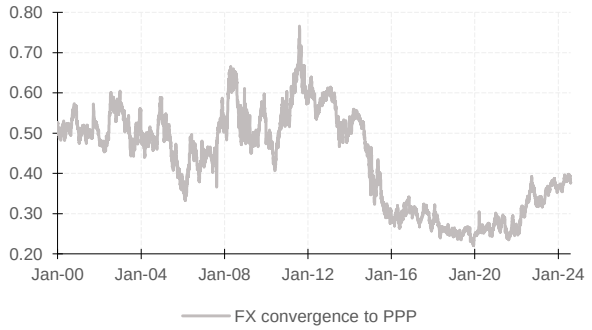
Model based on rates expectations current positioning



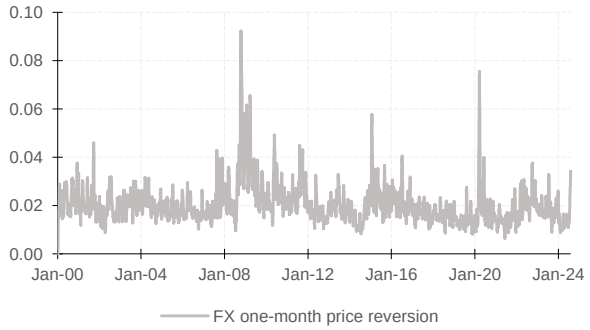
Aggregated current positioning



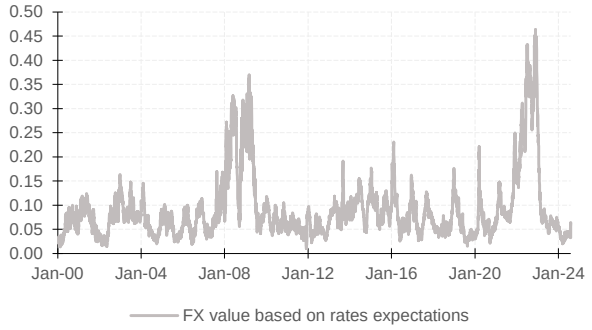
PPP-based model – Historical signal strength



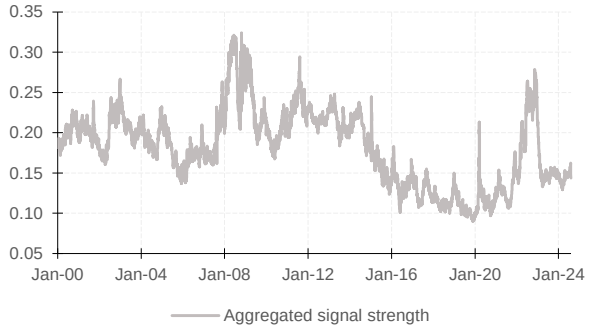
Mean reversion-based model – Historical signal strength



Rates expectations-based model – Historical signal strength



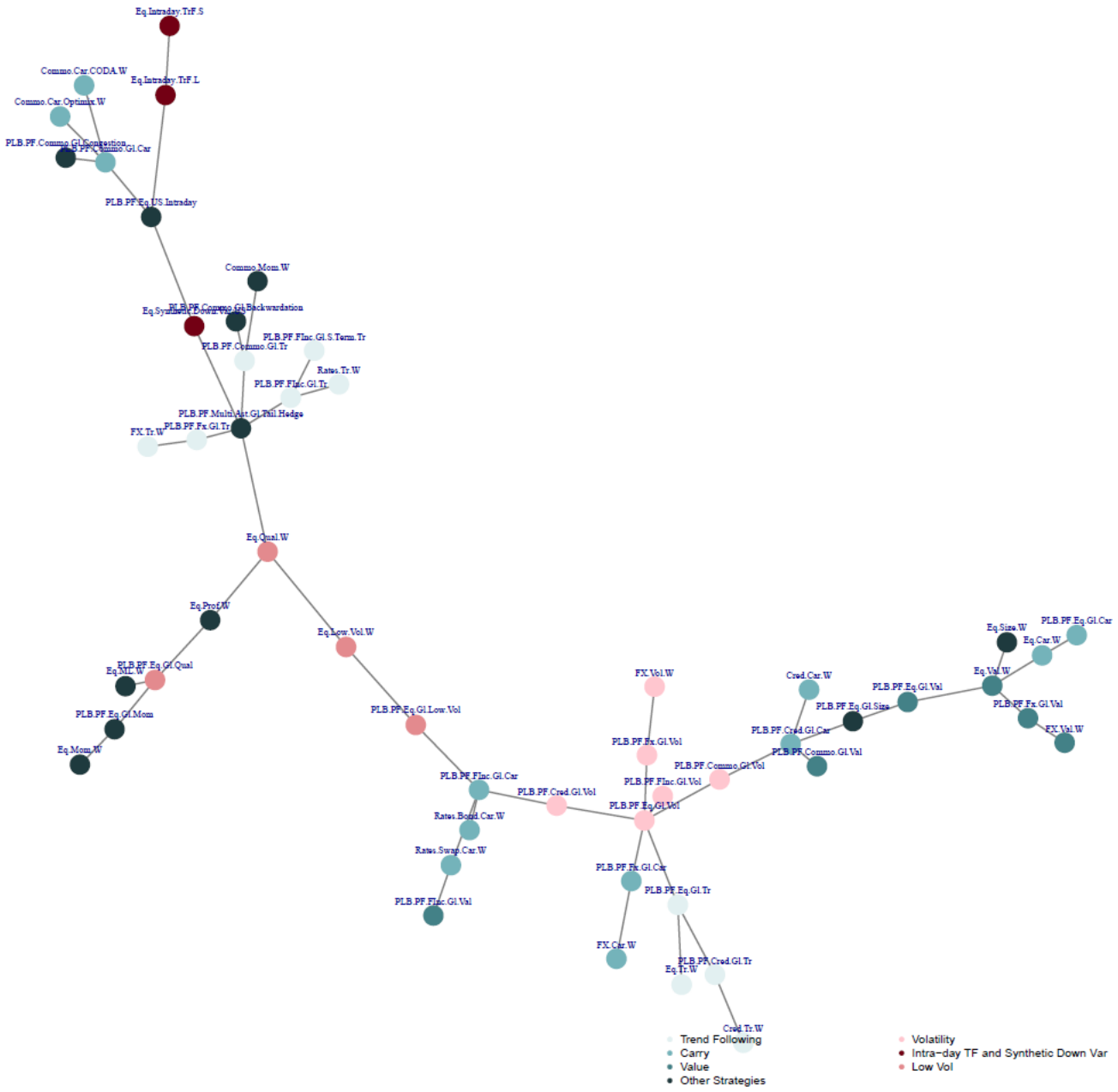
Aggregated current positioning – Historical signal strength



Source: SG Cross Asset Research. The historical signal strength is the difference between the average of the top three signals and the average of the bottom three signals, calculated historically.

Correlation analysis

A constellation of systematic strategies[1]

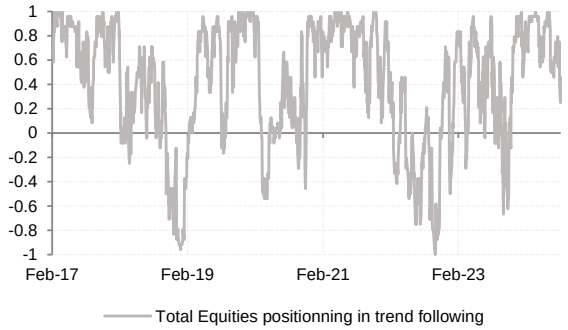


Source: SG Cross Asset Research/Cross Asset Quant, SGI, Bloomberg

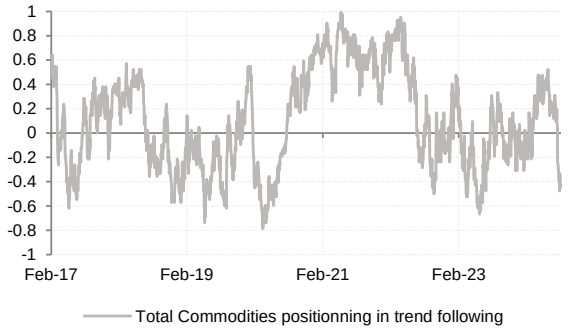
[1] The graph above is a spanning tree. A spanning tree identifies the strategy that is most correlated with the others and puts it in the middle of the graph. It then proceeds to build a number of branches so that the highly correlated strategies appear next to one another in ever decreasing order of correlation. It is an alternative way to visualise correlations among a universe of assets or strategies. See the Addendum for more details.

Strategy positioning – Cross asset trend following

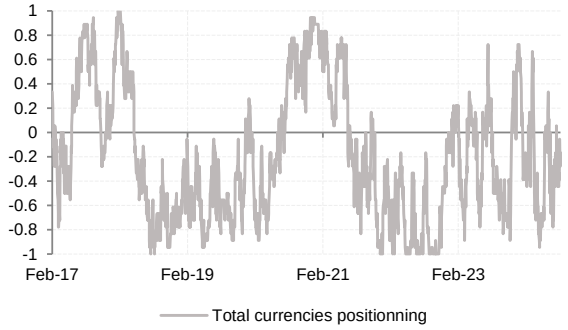
Total equities positioning



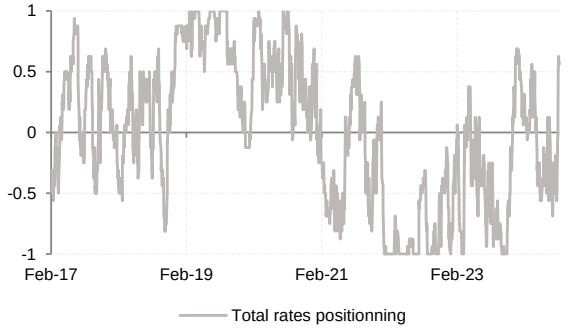
Total commodities positioning



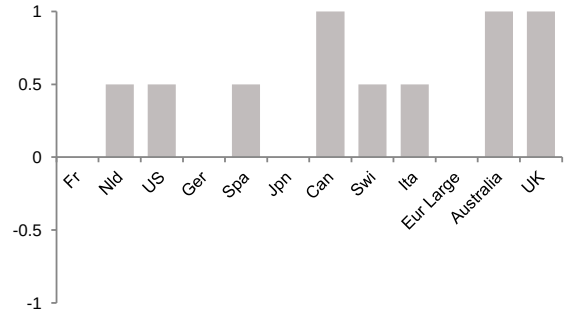
Total currencies positioning



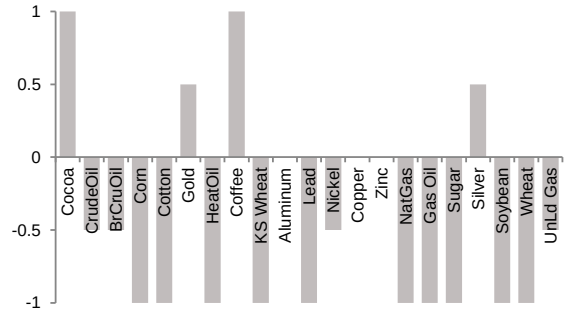
Total rates positioning



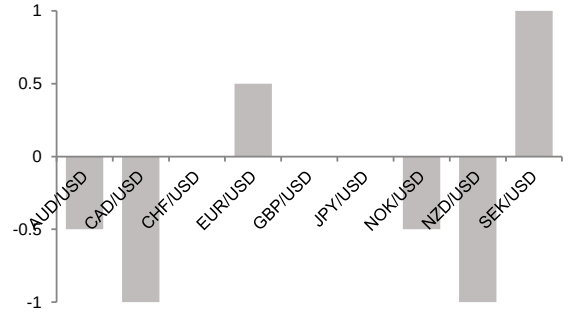
Equities current positioning



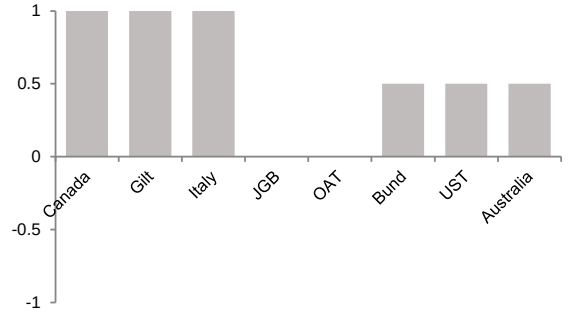
Commodities current positioning



FX current positioning



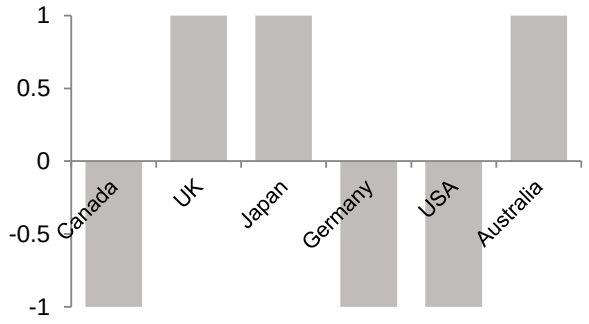
Rates current positioning



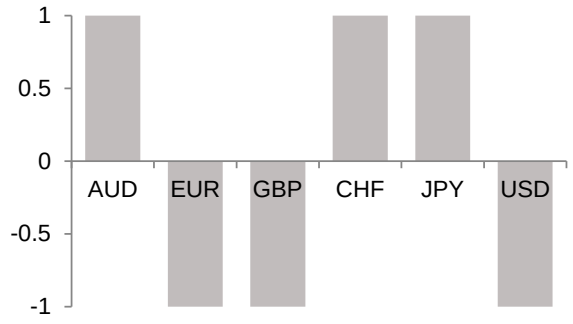
Source: SG Cross Asset Research/Cross Asset Quant

Strategy positioning – Bond and swap carry

Bond carry positioning



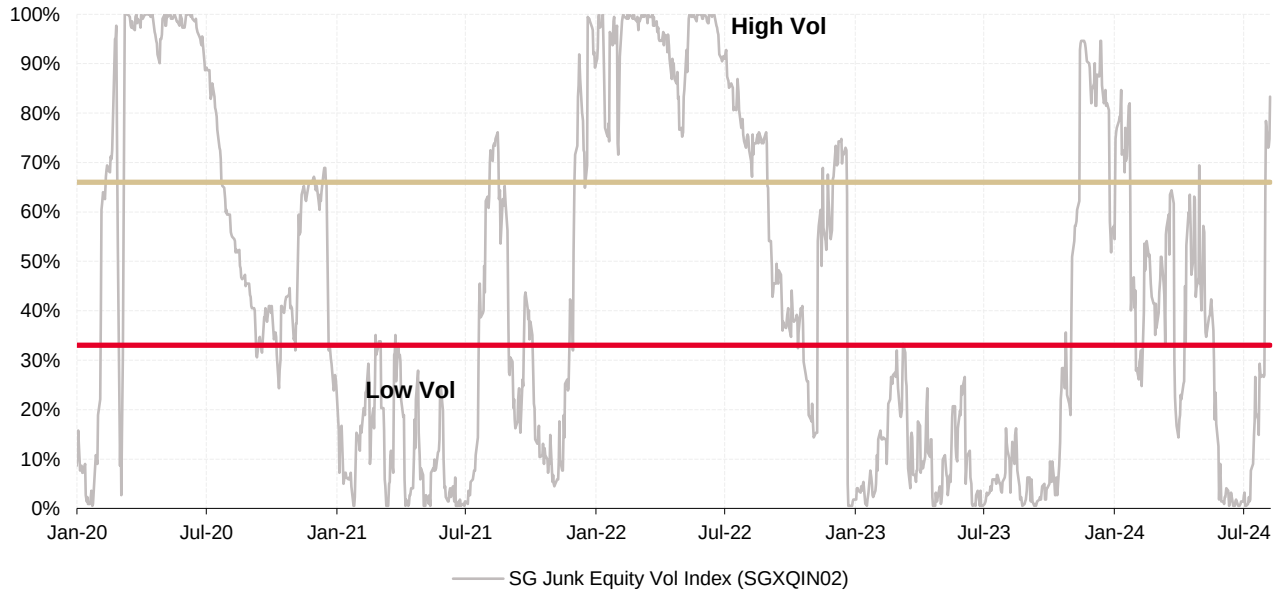
Swap carry positioning



Source: SG Cross Asset Research/Cross Asset Quant

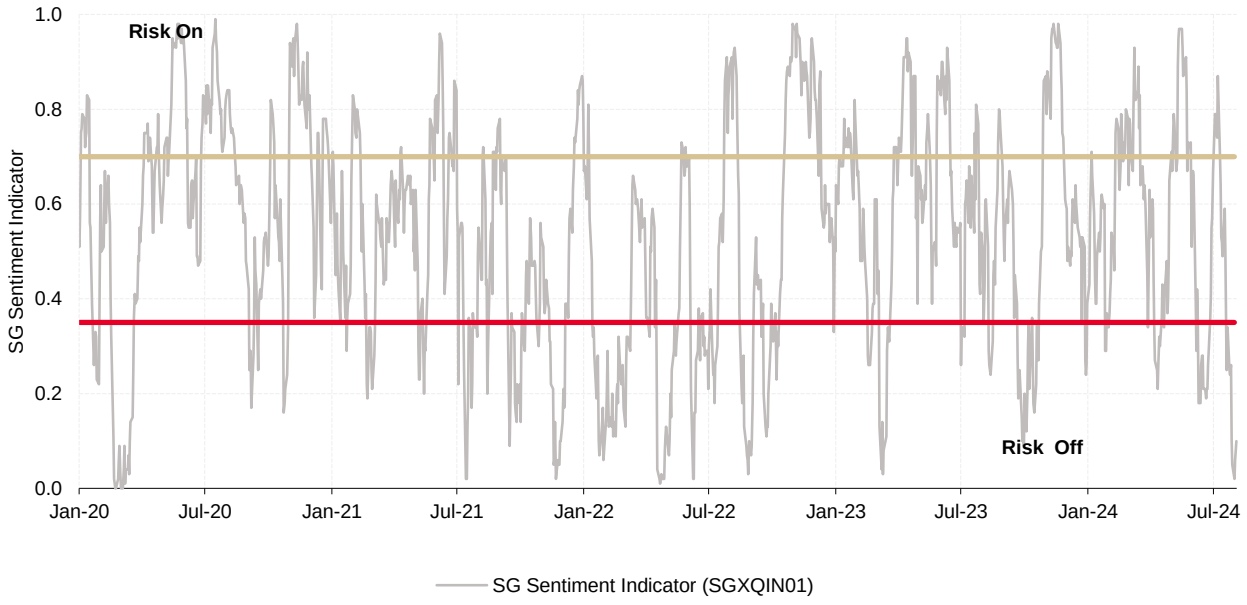
Tactical indicators

Junk equity volatility



Source: SG Cross Asset Research/Cross Asset Quant

SG Sentiment Indicator



Source: SG Cross Asset Research/Cross Asset Quant

Addendum

Detail of the statistics calculated on pages 2 and 3.

Year-to-date Return / Volatility / (Return/Risk)
All year-to-date (YTD) calculations start from the first business day of January of the current year if the current month is later than February, i.e. March and onwards. If the current month is February or sooner, YTD calculations start from the first business day of January of the previous year.

Tail correlation:
We look at the days where the returns of the benchmark (MSCI World and US 10y Treasuries in our case) are in their 5% and 95% percentile. On those days, we calculate the Spearman's rank correlation between the benchmark and the strategies.

Detail of the statistics used on page 6.

Correlation analysis:
In the spanning tree, to avoid using overlapping returns to calculate correlation, we use a robust method: the correlation is calculated on weekly returns that are separated into five non-overlapping partitions: Monday to Monday returns, Tuesday to Tuesday returns and so on. Then, correlation is calculated on each partition separately and aggregated as an average of the five partitions.

In terms of the equation, robust correlation is calculated as follows:

$$RobustCorr(A,B) = \frac{1}{5}\sum_{i=1}^{5}\frac{1}{\frac{N}{5}-2}\left(\sum_{j=0}^{\frac{N}{5}-2}(Ret^{A}_{5*j+i} - mean(Ret^{A}_{5*j+i}))(Ret^{B}_{5*j+i} - mean(Ret^{B}_{5*j+i}))\right)$$

where $Ret^{A}_{5*j+i} = \frac{S^{A}_{5*j+i}}{S^{A}_{5*(j+1)+i}} - 1$; S^{A}_{5*j+i} is the index level, and N is the number of days

References

Strategies used in this note have been detailed in past research papers.

Mapping of strategies and our research papers

	Paper	Publication date
Curve carry in rates	Rolling up the EUR swap curve	July 2013
Equity Value	'Patient versus brave' value investors - Introducing our NEW global value index	March 2014
Credit Trend	Credit momentum strategy	May 2014
Commodities carry	Risk premia investing - Introducing our flagship risk premia portfolio	October 2014
Equity Quality, Low Vol, Profitability, Momentum	Can top-down factor investing be a more profitable way to invest in equities?	October 2015
Credit Compression	Risk premia outlook: Market uncertainty justifies a risk premia approach	February 2016
Equity Tail Hedge, Synthetic down var	Hedges are not a luxury	October 2016
Cross Asset Trend	Why you need trend systems in your portfolio	June 2016
Duration hedge	Fixed Income Portfolio Strategy - The worst is yet to come	February 2017
Mean reversion	Benefiting from fluctuations in markets	March 2017
Cross Asset Volatility	A close-up view of the option premium	September 2017
Rates, FX carry and credit compression	Carry strategy in rates markets	September 2018
Intraday Trend-Following	Short-term trend following – How to monetise intraday trading patterns	April 2019
Tail risk hedge with equity and credit	Convexity for the Ice Age - Systematic credit and equity option strategies	September 2019
Value in Rates, FX and Commodity	Value investing in fixed income, currencies and commodities	September 2019

Source: SG Cross Asset Research/Cross Asset Quant

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